

**PART-I****S.No:** B304090**Programme** : B.Tech**Hall Ticket No:** 24B11AI239**Name** : Mamidi Amruthavalli Supriya**Examination** : III SEMESTER END EXAMINATIONS

REGULAR

Month-Year : Nov-25**Branch** : Artificial Intelligence &
Machine Learning**Course Code** : 241MA009**Course Name** : Probability & Statistics**Date of Exam** : 18.11.2025

Signature of the Controller of Examination

M.A.V. Supriya 18-11-2025
Signature of the Student with date

Signature of the Invigilator with date

Serial No.
Last Page Written

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INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
 - i. Writing their Hall Ticket No. and name in any part of the answer booklet.
 - ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
 - iii. Writing religious symbols.
 - iv. Either seeking or providing any assistance to the fellow students in the examinations.
 - v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
 - vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
 - vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**



SPACE FOR ROUGH WORK

M. V. V. Subrah 12-11-2012

12/11/12

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2a) Given

$$f(x) = kx^2 e^{-x}$$

To find value of k

W.K.T

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx = 1$$

$$0 + \int_0^{\infty} kx^2 e^{-x} dx = 1$$

$$k \int_0^{\infty} x^2 e^{-x} dx = 1$$

$$k [x^2(-e^{-x}) - 2x(e^{-x}) + 2(-e^{-x})]_0^{\infty} = 1$$

$$k [0 - 0 + 2[-(0-1)]] = 1$$

$$k [2(1-0)] = 1$$

$$2k = 1$$

$$k = \frac{1}{2}$$

$$\text{Mean} = \int_0^{\infty} x \cdot f(x) dx$$

$$= \int_0^{\infty} x \cdot kx^2 e^{-x} dx$$

$$= k \int_0^{\infty} x^3 e^{-x} dx$$

$$= k [x^3(-e^{-x}) - 3x^2(e^{-x}) + 6x(-e^{-x}) - 6(e^{-x})]_0^{\infty}$$

$$= k [0 - 0 + 0 - 6[0-1]]$$

$$= k [6]$$

$$= \frac{6}{2}$$

$$= 3$$

$$\therefore \text{Mean} = 3$$



$$\text{Variance} = \int_0^{\infty} x^2 f(x) dx - \mu^2$$

$$= \int_0^{\infty} x^2 \cdot k \cdot x^2 e^{-x} - \mu^2$$

$$= k \int_0^{\infty} x^4 e^{-x} - \mu^2$$

$$= k [x^4(-e^{-x}) - 4x^3(-e^{-x}) + 12x^2(-e^{-x}) - 24x(e^{-x}) + 24(-e^{-x})]_0^{\infty} - \mu^2$$

$$= k [0 - 0 + 0 - 0 + 24[-(0-1)]] - \mu^2$$

$$= k [24] - \mu^2$$

$$= \frac{1}{2} [24] - \mu^2$$

$$= 12 - (3)^2$$

$$= 12 - 9$$

$$= 3$$

$$\therefore \text{variance} = 3$$

2b) Given

$$f(x) = \begin{cases} 0 & \text{if } x \leq 1 \\ k(x-1)^3 & \text{if } 1 \leq x \leq 3 \\ 1 & \text{if } x > 3 \end{cases}$$

i) To find k

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^0 0 dx + \int_1^3 f(x) dx + \int_3^{\infty} f(x) dx = 1$$

$$\int_1^3 k(x-1)^3 dx + \int_3^{\infty} 1 dx = 1$$

$$k \int_1^3 (x-1)^3 dx + (-4) = 1$$



$$K \int_1^3 x^3 dx + \int_1^3 1 dx - \int_1^3 3x^2 dx - \int_1^3 3x dx = 5$$

$$K \left[\left[\frac{x^4}{4} \right]_1^3 + [x]_1^3 - 3 \left[\frac{x^3}{3} \right]_1^3 - 3 \left[\frac{x^2}{2} \right]_1^3 \right] = 5$$

$$K \left[\left[\frac{81}{4} - \frac{1}{4} \right] + [3-1] - 3 \left[3^3 - 1^3 \right] - \frac{3}{2} [3^2 - 1^2] \right]$$

$$K \left[\frac{1}{4} [80] + 2 - [27-1] - \frac{3}{2} [8] \right]$$

$$K [20 + 2 - 26 - 12] = 5$$

$$K [22 - 38] = 5$$

$$K [-16] = 5$$

$$K = -5/16$$

ii) Pdf of x .

$$\text{Given } f(x) = \int_{-\infty}^{\infty} f(x) dx$$

$$= \int_{-\infty}^0 f(x) dx + \int_0^3 f(x) dx + \int_3^{\infty} f(x) dx$$

$$= 0 + \int_0^3 K(x-1)^3 dx + 1 = \frac{5}{16} \int_0^3 (x-1)^3 dx + 1$$

$$\text{iii) mean } \mu = \int_1^3 x (f(x) dx)$$

$$= \int_1^3 K x (x-1)^3 dx$$

$$= \int_1^3 K (x^2 - x)^3 dx$$

$$= K \left[\int_1^3 x^6 dx + \int_1^3 x^3 dx - 3 \int_1^3 x^5 dx - 3 \int_1^3 x^4 dx \right]$$

$$= K \left[\left[\frac{x^7}{7} \right]_1^3 + \left[\frac{x^4}{4} \right]_1^3 - 3 \left[\frac{x^6}{6} \right]_1^3 - 3 \left[\frac{x^5}{5} \right]_1^3 \right]$$

$$= K \left[\left[\frac{3^7}{7} - \frac{1}{7} \right] + \left[\frac{3^4}{4} - \frac{1}{4} \right] - \frac{3}{6} \left[\frac{3^6}{6} - \frac{1}{6} \right] - \frac{3}{5} [3^5 - 1] \right]$$

$$= K \left[\frac{1}{7} [2187 - 1] - \frac{3}{6} [729 - 1] - \frac{3}{5} [243 - 1] \right]$$

$$= K [312.28 - 364 - 145.8]$$

$$= K [-197.52] = -5/16 (-197.52) = 6.1725$$



3a)

Given

$$n = 1000$$

$$\mu = 78$$

$$\sigma = 11$$

i) Students got

$$z = \frac{\bar{x} - \mu}{\sigma}$$

$$z = \frac{\bar{x} - 78}{11}$$

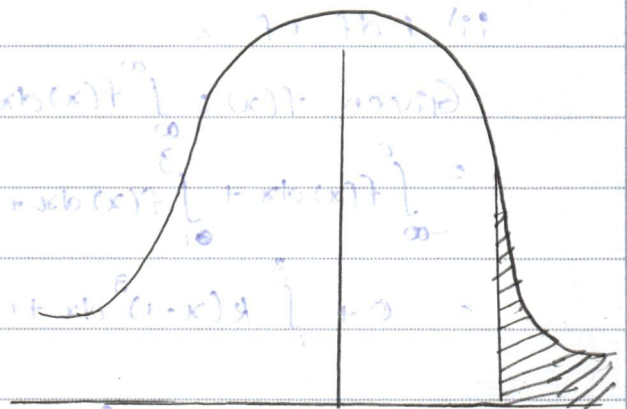
ii) Students got marks above 90

$$P(x > 90) =$$

$$z = \frac{90 - 78}{11}$$

$$= \frac{12}{11}$$

$$= 1.09$$



$$P(x > 90) = P(z > 1.09)$$

$$= P(z > 1.09)$$

$$= 0.5 - A(1.09)$$

$$= 0.5 - 0.3621$$

$$= 0.1379$$

No. of students got marks greater than 90 is

$$= 1000(0.1379)$$

$$= 137.9$$

$$\approx 138 \text{ students}$$



ii) Students got marks below 60

$$P(X < 60)$$

$$Z = \frac{60 - 78}{11} = \frac{-18}{11} = -1.636$$

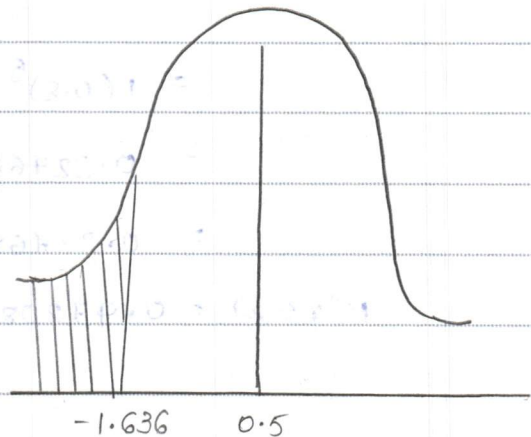
$$P(X < 60) = P(Z < -1.636)$$

$$= 0.5 - A(Z)$$

$$= 0.5 - A(1.636)$$

$$= 0.5 - 0.4484$$

$$= 0.0516$$



No. of students got marks

$$\text{less than } 60 = 1000(0.0516)$$

$$= 51.6 \approx 51$$

3b) Given sample chosen at random $(n) = 5$

$$\text{There are 20% defective } (P) = \frac{20}{100} = 0.2$$

$$\therefore P + q = 1$$

$$q = 1 - P = 1 - 0.2 = 0.8$$

$$P(X = r) = {}^n C_r P^r q^{n-r}$$

i) none is defective

$$P(\bar{X} = 0) \Rightarrow {}^n C_r P^r q^{n-r}$$

$$= {}^5 C_0 (0.2)^0 (0.8)^{5-0}$$

$$= 1(0.2)^0 (0.8)^5$$

$$= 1(0.8)^5$$

$$P(X = 0) = 0.04 \quad 0.32768$$

ii) one is defective

$$P(\bar{X} = 1) \Rightarrow {}^n C_r P^r q^{n-r}$$

$$= {}^5 C_1 (0.2)^1 (0.8)^{5-1}$$

$$= 5(0.2)(0.8)^4 = 0.4096$$



iii) At most 2 defective

$$P(r \leq 2) = P(r=0) + P(r=1) + P(r=2)$$

$$= {}^5C_0 (0.2)^0 (0.8)^5 + {}^5C_1 (0.2)^1 (0.8)^{5-1} + {}^5C_2 (0.2)^2 (0.8)^{5-2}$$

$$= 1(0.8)^5 + 5(0.2)(0.8)^4 + 10(0.04)(0.8)^3$$

$$= 0.32768 + 0.4096 + (0.4)(0.512)$$

$$= 0.32768 + 0.4096 + 0.2048$$

$$P(r \leq 2) = 0.94208$$

5a) Data:

- Data can store numbers, variables, values etc...
- Data can be classified into many types mainly classified into two types

i) Qualitative data

ii) Quantitative data

i) Qualitative data:

- Data is stored as non-numeric. i.e. can't store numerical data
- Stores data like characteristics etc..

This quantitative data is classified into two types.

i) Normal data:

- Data is not stored in specific way

ii) Ordered data:

- Data is stored in specific ordered way.



i) Quantitative data:

- Numerical data is stored in this datatype.
i.e., height, weight, row numbers etc...

Quantitative data is classified into two subtypes

i) Discrete data:

- Data must be countable i.e., 1, 2, 3...

ii) Continuous data:

- Data is continuous at interval of times.

There are different types of or methods of data collection. Most important are

i) primary data collection

ii) Secondary data collection.

i) primary data collection:

- Data is directly collected from source or person with directly interacting.

Methods:

- Interviews : conducting interviews
- Surveys : By conducting surveys to get information
- Questionnaires : Face to face interactions or asking questions. etc...

ii) Secondary data collection:

- Data is collected from external source
i.e., collect data indirectly.

Methods:

- Books



Q.No.

- Newspapers

- Social media etc...

5b. Given data

2, 3, 6, 8, 11

$N = 5$; given sample size $(n) = 2$

i) Mean of population.

$$\mu = \frac{\text{Sum of all observations}}{\text{No. of all observations}}$$

$$= \frac{2+3+6+8+11}{5}$$

$$= \frac{30}{5}$$

$$\boxed{\mu = 6}$$

ii) Standard deviation of population.

$$\text{variance } (\sigma^2) = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}$$

$$= \frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{5}$$

$$= \frac{4^2 + 3^2 + 0^2 + 2^2 + 5^2}{5}$$

$$= \frac{16 + 9 + 0 + 4 + 25}{5}$$

$$= \frac{54}{5}$$

$$= 10.8$$

$$\text{Standard deviation } \sigma = \sqrt{10.8} = 3.28$$

$$\boxed{\sigma = 3.28}$$

- Book



The 25 outcomes are,

(2, 3), (2, 6), (2, 8), (2, 11), (2, 2)

(3, 6), (3, 8), (3, 11), (3, 2), (3, 3)

(6, 8), (6, 11), (6, 2), (6, 3), (6, 6)

(8, 11), (8, 2), (8, 3), (8, 6), (8, 8)

(11, 2), (11, 3), (11, 6), (11, 8), (11, 11)

now, The Mean of the sampling distribution of Means is

$$\Rightarrow 2.5 + 4 + 5 + 6.5 + 2 + 4.5 + 5.5 + 7 + 2.5 + 3 + 7 + 8.5 + 4 + 4.5 + 6 + 9.5 + 5 + 5.5 + 7 + 8 + 6.5 + 7 + 8.5 + 9.5 + 11$$

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$$\Rightarrow 5.6$$

iv) the standard deviation of the sampling distribution of Means

$$\sigma^2 = (2.5 - 5.6)^2 + (4 - 5.6)^2 + (5 - 5.6)^2 + (6.5 - 5.6)^2 + (2 - 5.6)^2 + (4.5 - 5.6)^2 + (5.5 - 5.6)^2 + (7 - 5.6)^2 + (2.5 - 5.6)^2 + (3 - 5.6)^2 + (7 - 5.6)^2 + (8.5 - 5.6)^2 + (4 - 5.6)^2 + (4.5 - 5.6)^2 + (6 - 5.6)^2 + (9.5 - 5.6)^2 + (5 - 5.6)^2 + (5.5 - 5.6)^2 + (7 - 5.6)^2 + (8 - 5.6)^2 + (6.5 - 5.6)^2 + (7 - 5.6)^2 + (8.5 - 5.6)^2 + (9.5 - 5.6)^2 + (11 - 5.6)^2$$

25



iv) The standard deviation of the sampling distribution of Mean.

$$\Rightarrow \frac{(2.5-6)^2 + (4-6)^2 + (5-6)^2 + (6.5-6)^2 + (2-6)^2 + (4.5-6)^2 + (5.5-6)^2 + (7-6)^2 + (2.5-6)^2 + (3-6)^2 + (7-6)^2 + (8.5-6)^2 + (4-6)^2 + (4.5-6)^2 + (6-6)^2 + (9.5-6)^2 + (5-6)^2 + (5.5-6)^2 + (7-6)^2 + (8-6)^2 + (6.5-6)^2 + (7-6)^2 + (8.5-6)^2 + (9.5-6)^2 + (11-6)^2}{25}$$

$$\Rightarrow \frac{12.25 + 4 + 1 + 0.25 + 16 + 2.25 + 0.25 + 1 + 12.25 + 9 + 1 + 6.25 + 4 + 2.25 + 0 + 12.25 + 1 + 0.25 + 1 + 4 + 0.25 + 1 + 6.25 + 12.25 + 25}{25}$$

$$\Rightarrow 5.4$$

$$\text{Hence, } \sigma^2 = 5.4 \Rightarrow \sigma = \sqrt{5.4}$$

$$\Rightarrow \text{a) Given, } n = 49, \quad \bar{x} = 15200$$

$$\mu = 15150$$

$$\sigma = 120$$

$$\alpha = 0.05$$

Now, From the given data we can observe that $n > 30$ we consider the z-test

Now, we have to find the value of $\frac{z_{\alpha}}{2} = 1.96$

The critical Test $\frac{z_{\alpha}}{2} = 1.96$



Here,

Null Hypothesis $\mu_0 = \mu$

Alternative Hypothesis $\mu_0 \neq \mu$

From,

Test significance $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

$$z = \frac{15200 - 15150}{120/\sqrt{49}}$$

$$z = \frac{50}{\frac{120}{7}}$$

$$z = \frac{50}{17.14}$$

$$z = 2.91$$

Hence, the calculated value of

$$z_{cal} = 2.91$$

The Tabulated Value of z is 1.96

Here, $z_{cal} > z_{Tabulated}$

The Null Hypothesis was rejected in this case.



7) b) Given, $n_1 = 6400$

$$n_2 = 1600$$

$$\bar{x}_1 = 67.85$$

$$\bar{x}_2 = 68.55$$

$$\sigma_1 = 2.56$$

$$\sigma_2 = 2.52$$

$$\text{and } \alpha = 0.01$$

Here we consider the z-Test for this Condition Because the both $(n_1, n_2 > 30)$

The Test Significance Formula

$$i) z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Now, The critical Test $\frac{2\alpha}{2} = 1.96$

The Test Significance.

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\Rightarrow z = \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} + \frac{(2.52)^2}{1600}}}$$

$$\Rightarrow z = \frac{-0.7}{\sqrt{\frac{6.55}{6400} + \frac{6.35}{1600}}}$$

$$\Rightarrow z = \frac{-0.7}{\sqrt{1.023 + 3.968}} \Rightarrow \frac{-0.7}{\sqrt{4.99}} \Rightarrow \frac{-0.7}{2.23}$$

$$\Rightarrow \frac{-0.7}{\sqrt{4.99}} \Rightarrow \frac{-0.7}{2.23}$$



$$= -0.313$$

Here, $z_{cal} = -0.313$

Here,

$$z_{tab} > z_{cal}$$

Hence the Null Hypothesis was accepted

10)a) Given data,

Height of father (x)	Height of son (y)	X ($x - \bar{x}$)	Y ($y - \bar{y}$)	XY	$x+y$
65	67	-3.25	-3.5	11.3	132
66	68	-2.25	-2.5	5.62	134
67	64	-1.25	-1.5	1.87	131
67	68	-1.25	-1.5	1.87	135
68	72	-0.25	-0.5	0.125	140
69	70	0.75	0.5	0.375	139
71	69	2.75	2.5	6.875	140
73	70	4.75	4.5	21.37	143

Karl Pearson's coefficient of Correlation

$$r = \frac{\sum XY}{\sqrt{\sum x^2 \sum y^2}}$$

Here

$$x = x - \bar{x}$$

$$y = y - \bar{y}$$



$$\bar{x} = \frac{65 + 66 + 67 + 67 + 68 + 69 + 71 + 73}{8}$$

$$= 68.25$$

$$\bar{y} = \frac{67 + 68 + 64 + 68 + 72 + 70 + 69 + 70}{8}$$

$$= 68.5$$

$$\therefore \Sigma xy = 49.735$$

$$\Sigma x + y = 4.$$

Hence, The Karl Pearson's Coefficient of correlation is $\frac{49.73}{4}$.

10)b) Given, $3x + 12y = 19, \rightarrow (1)$

$$3y + 9x = 46 \rightarrow (2)$$

The two regression lines

are $y = ax + byx$

$$x = ax + byx$$

Now, From (1) $\Rightarrow$

$$3x + 12y = 19$$

$$12y = 19 - 3x$$

$$y = \frac{19 - 3x}{12}$$

$$\Rightarrow y = \frac{19}{12} - \frac{3}{12}x$$

Here, $byx = -\frac{3}{12}$

From (2) $\Rightarrow$

$$3y + 9x = 46$$

$$9x = 46 - 3y$$

$$x = \frac{46 - 3y}{9}$$



1a) Given Three Bags

Bag 1 : 1 white, 2 red, 3 green - 6 balls

Bag 2 : 2 white, 3 red, 1 green - 6 balls

Bag 3 : 3 white, 1 red, 2 green - 6 balls.

Two balls are taken at random from any bag.

$$P(B_i/Ball) = 2/6.$$

$$P(W) = 0.5$$

$$P(R) = 0.5$$

$$P(W/B_1) =$$

$$P(Bag/Ball) =$$

$$P(B_1/W) = 1/6$$

$$P(B_1/R) = 2/6$$

$$P(B_2/W) = 2/6$$

$$P(B_2/R) = 3/6$$

$$P(B_3/W) = 3/6$$

$$P(B_3/R) = 1/6.$$

Balls are drawn from Bag 2

$$P(W/B_2) \cup P(R/B_2) = P(W/B_2) + P(R/B_2)$$

$$P(W/B_2) = \frac{P(W) \cdot P(B_2/W)}{P(W) \cdot P(B_2/W) + P(W) \cdot P(B_1/W) + P(W) \cdot P(B_3/W)}$$

$$= \frac{(0.5)(2/6)}{(0.5)(2/6) + (0.5)(1/6) + 0.5(3/6)}$$

$$= \frac{0.5(2/6)}{0.5 \left[\frac{2}{6} + \frac{1}{6} + \frac{3}{6} \right]}$$

$$= \frac{0.5(2/6)}{0.5 \left[\frac{2}{6} + \frac{1}{6} + \frac{3}{6} \right]}$$

$$= \frac{2/6}{\frac{2}{6} + \frac{1}{6} + \frac{3}{6}}$$

$$= \frac{2/6}{6/6}$$

$$= 2/6$$

$$= \frac{1}{3}$$

$$= 0.333$$



Q.No.

$$\begin{aligned}
 P(R/B_2) &= \frac{P(R) \cdot P(B_2/R)}{P(R) \cdot P(B_2/R) + P(R) \cdot P(B_1/R) + P(R) \cdot P(B_3/R)} \\
 &= \frac{P(R) \cdot P(B_2/R)}{P(R) (P(B_1/R) + P(B_2/R) + P(B_3/R))} \\
 &= \frac{P(B_2/R)}{P(B_1/R) + P(B_2/R) + P(B_3/R)} \\
 &= \frac{3/6}{3/6 + 2/6 + 1/6} \\
 &= \frac{3/6}{\frac{2+3+1}{6}} \\
 &= \frac{3}{6} \\
 &= \frac{1}{2} \\
 &= 0.5
 \end{aligned}$$

probability of balls taken from B_2

$$\begin{aligned}
 &P(B_1/B_2) + P(R/B_2) \\
 &= 0.33 + 0.5 \\
 &\approx 0.83
 \end{aligned}$$

1b) Given

x	0	1	2	3	4	5	6	7
$P(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

i) K

$$\sum_{i=0}^7 P_i = 1$$

$$0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$$



$$10K^2 + 9K = 1$$

$$10K^2 + 9K - 1 = 0$$

$$K = \frac{-9 \pm \sqrt{81 - 4(10)(-1)}}{2(10)}$$

$$= \frac{-9 \pm \sqrt{81 + 40}}{20}$$

$$= \frac{-9 \pm \sqrt{121}}{20}$$

$$= \frac{-9 \pm 11}{20}$$

$$= \frac{-9 + 11}{20} \quad \text{or} \quad \frac{-9 - 11}{20}$$

$$= \frac{2}{20} \quad \text{or} \quad \frac{-20}{20}$$

$$= \frac{1}{10} \quad \text{or} \quad -1$$

$$K = 0.1$$

$$K = 0.1$$

neglect negative

$$P(X < 6) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= 0 + K + 2K + 2K + 3K + K^2$$

$$= 8K + K^2$$

$$= 8(0.1) + (0.1)^2$$

$$= 0.8 + 0.01$$

$$= 0.81$$

$$P(0 < X < 5) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

$$= K + 2K + 3K + 2K$$

$$= 8K$$

$$= 8(0.1) = 0.8$$

$$= 0.8$$



iv) mean

$$\mu^* = \sum_{i=0}^7 x_i p_i$$

$$= k + 2(2k) + 3(2k) + 4(3k) + 5(k^v) + 6(2k^v) + 7(7k^v + k)$$

$$= k + 4k + 6k + 12k + 5k^v + 12k^v + 49k^v + 7k$$

$$= 66k^v + 30k$$

$$= 66(0.1)^v + 30(0.1)$$

$$= 66(0.01) + 30(0.1)$$

$$= 0.66 + 3$$

$$= 3.66$$

v) Variance

$$\sum_{i=0}^7 x_i^2 p_i$$

$$= 1(k) + 4(2k) + 9(2k) + 16(3k) + 25(k^v) + 36(2k^v) + 49(7k^v + k)$$

$$= k + 8k + 18k + 48k + 25k^v + 72k^v + 343k^v + 49k$$

$$= 440k^v + 124k$$

$$= 440(0.01) + 124(0.1)$$

$$= 4.4 + 12.4$$

$$= 16.8$$

$$\sigma^2 = 16.8$$

or

$$\text{variance} = \sum_{i=0}^7 x_i^2 p_i - \mu^2$$

$$= 16.8 - (3.66)^2$$

$$= 16.8 - 13.39$$

$$= 3.41$$





Q.No.





Q.No.



Q.No.



Q.No.



Q.No.



Q.No.





Q.No.



Q.No.



Q.No.





Q.No.



